01 / GigBid engineering review

AI-4016 / Raahim / November 2026

A freelancer copilot with a durable human approval boundary, portfolio-grounded drafts, and an independently implemented student marketplace simulator.

This engineering review distinguishes implemented behaviour, controlled synthetic measurements and evidence still missing. It is not a claim of completed hidden-test performance or a deployed public service. Negative model results are retained.

Problem and constraints

The workflow is discovery, risk and fit assessment, pricing, grounded drafting, human review and client response. Automation may recommend, skip or prepare an action. It must never send a proposal, send a message or spend Connects without a matching approval. Edits invalidate the previous approval and are rechecked.

Declared budgets

The configured targets are p95 analysis within 60 seconds, proposal preparation within 120 seconds, and inference API targets of USD 0.02 and USD 0.05 respectively. Thread budgets bound steps, tokens and Critic revisions. Targets are not achieved measurements. Local API charges exclude GPU hardware and electricity.

Evidence class	Observed scope
Backend control/integration	214 passing checks at the recorded checkpoint; opt-in runtime tests are separate.
Browser flows	Existing flows plus a separate simulator outage/retry/submission/reply test; deterministic role fixtures.
Retrieval study	60 answerable synthetic tasks plus six unrelated tasks; real BGE/MiniLM inference.
Local graph study	Two configurations, 30 synthetic jobs each. Grounding failures dominate.

Evidence: evals/results/engineering-verification.json; student-retrieval/results.json; student-generation-v3/progress.json. Baseline source checkpoint: 4fed02f.

02 / Architecture and ownership

The Next.js client uses typed FastAPI contracts. The API authenticates and persists work; Redis/Arq runs indexing and graph jobs. Postgres owns checkpoints, approval receipts, immutable effect records and the memory promotion ledger. Qdrant indexes tenant-filtered evidence IDs; source text remains in source storage.

Graph	Execution and rationale
Bid	Scout intake fans out to Analyst, Risk and Matcher. A join reconciles fit. Decision, pricing, writer, verifier, bounded Critic, compliance and exact preview lead to a human interrupt. An approved executor is the only write route.
Conversation	One durable thread per conversation. Liaison classifies intent and drafts using portfolio evidence. Calendar slots come from Workspace. Verification and a fresh interrupt precede a saved draft or simulator message.

Three separate stores

State contains current job/conversation facts and typed handoffs. Execution events record progress, usage and safe failures. Long-term memory contains only promoted observations: repeated human-confirmed edits for style, and verified provider events for outcomes. Raw tool output does not become memory.

Capability boundaries

Portfolio MCP exposes search, source reads, resources and a screening prompt. Marketplace MCP wraps one port. Workspace MCP provides local calendar and draft tools. The host binds tenant and run identity; these are not model arguments. The HTTP wrapper uses bearer authentication and a shared Redis limit. Actual Upwork OAuth and GitHub ingestion are still absent.

Crash and replay

The graph uses Postgres checkpoints, an advisory lock and a SQL worker-token fence. The latest approval binds interrupt ID, version, digest, tool, payload and idempotency key. The upstream simulator has an atomic debit/receipt transaction. Reconciliation records an already observed effect rather than authorising a new one. Calendar, budget and evidence freshness are rechecked at execution.

Evidence: agent/graphs; agent/runtime.py; mcp_servers; worker/simulator_sync.py; tests/integration/test_simulator_flow.py; docs/GRAPHS.md.

03 / Data and simulation

The instructor's follow-up allows approved Kaggle datasets and clearly identified synthetic fields. The original GigSim was not delivered; the user confirmed that students must build their own. Our implementation is independent, and hidden-test compatibility is not established.

The selected source is Asaniczka's Upwork Job Postings Dataset 2024, Kaggle version 1, with uploader-declared ODC Attribution terms. No Upwork website is scraped. Raw and derived job text is ignored by Git; attribution, version and hashes are documented. The database license does not independently establish rights in every job's text.

Source/preprocessing	Recorded count
Input CSV rows	53058
Selected per persona	200 AI/ML, 200 full-stack, 200 analyst
Missing/invalid budgets	8196
Text outside length bounds	101
Duplicate normalized content	23

Field provenance

Base titles, descriptions, publication dates and posted budgets are historical observations. Persona/skills, client identity/history, Connects costs, screening questions, inserted attacks and all outcomes are generated or derived. USD is an assumption; posted budgets are not human-validated fair-price labels. Inserted scam labels do not accuse the original job authors.

Fictional portfolios

Three separately labelled personas have 20 case studies, three duplicate READMEs, one CV and 150 generated past proposals each. These are not Raahim's experience. Generated historical won/replied labels are not experimental successes. The retrieval gold uses project identity and is easier than real-job relevance.

Client model

The baseline studies use a transparent rule client: eligibility, budget, length and skill coverage determine outcomes. Optional local LLM client mode has a durable outbox and bounded retries. Its outcomes are a different experimental condition and are not mixed into the rule-client results.

Evidence: docs/DATA_CARD.md; scripts/prepare_student_data.py; scripts/student_personas.py; evals/results/student-data-manifest.json; simulator/store.py and client.py.

04 / R&D B: measured retrieval

Predeclared H1: hybrid retrieval plus cross-encoder reranking improves Recall@5 over dense V0 on industry-specific tasks. The frozen cohort has identical tasks and project-identity gold across all variants. Embeddings are BGE-small; reranking uses MiniLM. Timing excludes the separately recorded model warm-up.

Variant	Recall@5	MRR@5	nDCG@10	p95 s	Unrelated retrieval
V0	1.000	0.987	0.990	0.050	6/6
V1	0.725	0.529	0.631	0.068	6/6
V2	0.983	0.858	0.894	0.061	6/6
V3	1.000	1.000	1.000	3.342	6/6
V8	1.000	1.000	1.000	10.715	0/6

Conclusion, shipped choice and ablations

H1 is unsupported: V0/V3 recall ties on this templated cohort. The declared recall, nDCG, latency rule selects V3 for ordering; V8 improves abstention at greater latency. V3/V2 ablates reranking, V2/V0 lexical fusion, V8/V3 correction. V1's generic requirements dilute retrieval. Per-persona rankings are retained; these measurements do not establish real-client reply benefits.

Grounding

Cover and screening claims need live supporting IDs; numerical achievements receive a source check. Source UUIDs are identifiers, not achievement numbers. Future plans have a non-factual verdict. Human calibration/panel appear on page 9.

V	Recall@5*	nDCG@10*	Bad / draft	Judge / 5	Reply / sent	p95/job s	USD/draft
V0	1.000	0.990	0.00 (n=5)	3.80	2/2	368.2	unknown
V3	1.000	1.000	0.00 (n=2)	4.00	1/1	124.4	unknown

*Retrieval uses the separate 60-answerable-task cohort. Proposal columns use nine matched jobs per variant with the same routed hosted open profile and 32,000-token caps. Every available draft is judged, including later failures. Intention-to-treat replies: V0 2/9, V3 1/9. V3's lower p95 includes more early failures, not a proven speed benefit. Small dev sample; billing unknown.

Before/after verification: all four changed checkpoint pairs were independently judged in a retrospective stratum. Unsupported factual claims were 2/22 (9.1%) before and 1/13 (7.7%) after; mean quality stayed 3.75/5. Across 54 completed passes the verifier removed four of 272 sentences. Nine unchanged control pairs received inconsistent factual labels, exposing judge noise. Failed/unpaired attempts are excluded from this conditional diagnostic, not counted as successes.

Evidence: evals/STUDENT_RESEARCH_PLAN.md; evals/results/student-retrieval/*.jsonl and results.json; rag-end-to-end-v1/analysis.json; verification-{changed,before-after}-v1/ under evals/results/; agent/proposal_nodes.py.

05 / Models and negative results

The frozen baseline compares all-role local Qwen 3.5 4B and Qwen 3 8B on the same 30 synthetic jobs. Retrieval is held at V0 and fallback is disabled. Errors and skips stay in the denominator. These are open-model configurations, not a comparison of orchestration frameworks. Source checkpoint 4fed02f preserves the baseline.

Configuration	Jobs	Statuses	p95 s	Candidates submitted
qwen3-8b	30	{'skip': 3, 'needs_attention': 26, 'watch': 1}	35.7	0
qwen3.5-4b	30	{'skip': 3, 'needs_attention': 27}	30.3	0

Interpretation

Most positive jobs stop at grounding. A diagnostic replay shows the local verifier calling future plans non-experience while still marking them unsupported. The writer also sometimes omits screening answers. This is a contract and prompt-handling failure as well as a model limitation. No useful reply or hire rate can be inferred from candidates that never reach submission.

Fix and hosted recheck

The revised contract distinguishes non-factual plans from past experience, explicitly tags the normal writer task, and keeps the past-claim guard. Groq GPT-OSS 20B/120B provide an optional hosted open-weight profile. A small control check correctly handles a future plan, unsupported number and supported source claim. The completed Gemini architecture study is separate from this frozen baseline; changed model, contract and retrieval prevent a clean causal model-only comparison.

Cost and speed

Usage records retain tokens, worker/model seconds and local generation throughput. Local inference API charge is zero but hardware/electricity are not priced. Qwen 4B generated 79.3 output tokens/s; 8B 64.4 tokens/s. The later inventory records an RTX 3070 Ti Laptop GPU (8 GB), Ryzen 9 6900HX and 15.2 GiB RAM; it is not an as-run hardware trace. Hosted actual billing is unknown unless verified, so it stays null. Groq has shared token/request guards and a daily study cap. A free API quota is not an unlimited budget, and rate-limit wait counts toward wall-clock latency.

Evidence: evals/results/student-generation-v3/plan.json, progress.json and raw model JSONL; local-hardware-inventory.json; agent/providers/groq.py; tests/unit/test_groq_models.py.

06 / Failure analysis

The table retains distinct failed jobs rather than selecting only successful proposals. Categories identify the visible stop condition; a stop does not prove that every removed sentence was actually false.

Job	Configuration	Category
student-4016-ai_ml-001	qwen3-8b	cover grounding
student-4016-ai_ml-002	qwen3-8b	cover grounding
student-4016-ai_ml-003	qwen3-8b	cover grounding
student-4016-ai_ml-004	qwen3-8b	cover grounding
student-4016-ai_ml-005	qwen3-8b	cover grounding
student-4016-ai_ml-006	qwen3-8b	cover grounding
student-4016-ai_ml-007	qwen3-8b	cover grounding
student-4016-ai_ml-008	qwen3-8b	cover grounding
student-4016-ai_ml-009	qwen3-8b	cover grounding
student-4016-full_stack-001	qwen3-8b	cover grounding

Screening grounding

Future validation plans were removed even when they made no past-work claim. The fix is an explicit not-applicable verdict for non-factual text, with a separate host guard that still rejects past experience mislabeled non-factual. Deterministic regressions pass; broader human validation remains necessary.

Cover grounding

Some drafts overstate demonstration work or name unproven libraries. Those sentences should remain blocked. Synthetic coursework is explicitly framed as a demonstration, source IDs stay in structured claims, and the ordinary writer task is now unambiguous. These changes must not erase baseline failures.

Schema and execution failures

Earlier diagnostic runs exposed missing study checkpoint initialisation and truncated verbose verifier output. Setup failures are separate from model-quality results. Durable storage is initialised before graph execution and verifier reasons are bounded by the prompt. No hidden success is inferred.

Evidence: evals/results/student-generation-v3/failures.json and raw records; tests/unit/test_proposal_nodes.py; docs/VERIFICATION.md.

07 / Safety, policy and operations

Approval is a transaction boundary

Read tools cannot grant write permission. The executor requires the exact approved preview and current draft digest. Changes to evidence, budget, calendar or connection can block execution. The simulator atomically deduplicates the effect and debit. Repeated worker calls do not spend twice.

Untrusted inputs

Job descriptions, portfolio text and client messages are data. Prompts delimit their role, and deterministic guards flag injected instructions, off-platform requests, contact details and guarantees. PII masking covers logs/events. The separate paired prompt-boundary experiment below measures raw role outputs, not end-to-end approval bypass or real-secret exposure.

Before/after injection boundary

GPT-OSS 120B processed 24 synthetic job/message attacks and four benign controls under each arm (56 calls). Before: 3/18 evaluable attacks disclosed the assigned fake marker/nonce; after: 4/18. Model errors were 6 and 6, excluded from the ASR denominators and retained in raw records. Existing role prompts form the baseline; the added shared boundary is the only intended change. Small designed samples and missing responses prevent a broad security claim; this is not a full-graph ablation. Sixteen fully observed pairs had no discordant outcomes (exact p=1.0); no improvement was established.

Policy update

Rules, model routes and thresholds are configuration. The announced instructor update was not supplied in this session. It would be dishonest to claim an observed change-request implementation time. The simulator's own rules are documented independently; they do not stand in for a missing instructor policy.

Production state

A temporary public HTTPS tunnel serves the full laptop-backed application and bearer-protected read-only Portfolio MCP. GitHub MCP imports public READMEs pinned to immutable commits. Three public LangSmith examples contain actual node/model/MCP spans with all source text omitted. URLs and anonymous-access verification are in README and public-traces.json. This deployment depends on the laptop remaining awake; it is not always-on cloud hosting. Real Upwork OAuth, encrypted credentials and live job reads are verified. Provider-preview writes are fixture-tested only; final submissions and direct messages are disabled. Production-wide auth limits remain gaps.

Recovery evidence

Backend and browser flows verify saved approval after reload, exactly-once simulator effects, an outage banner, explicit checkpoint retry, event import and approved replies. Browser role models are deterministic fixtures; those tests establish control behaviour rather than proposal quality.

Evidence: mcp_servers/marketplace/service.py; worker/simulator_sync.py; tests/integration/test_simulator_flow.py; apps/web/tests/e2e/simulator.spec.ts.

08 / R&D A/C: tools and architectures

Fifty identical fully synthetic jobs were run through a real ReAct agent, the langgraph-supervisor package and the production hand-built graph: 150 executions. Gemini 3.1/3.5 Flash-Lite assignment was blocked by job, balanced within persona, and identical across architectures. This optional closed-model study is separate from the two open-weight local configurations. Each run had a 20,000-token cap; native graphs had 30 steps and the hand graph 40, with two Critic revisions. The frozen plan, as-run source snapshot and all failed observations are retained.

Architecture	No error	Correct decision	Submitted	Replies	Tokens/job	p95 s
react	41/50	36/50	40/50	35/50	8377	138.6
supervisor	24/50	24/50	19/50	18/50	10330	186.1
hand_graph	37/50	49/50	30/50	29/50	8662	178.7

Decision versus completion

The hand graph's 49/50 correct decisions are intermediate decisions: 13 runs failed later. Error-free correct decisions were 36/50 for ReAct, 23/50 for supervisor and 36/50 for hand graph. The supervisor completed only 24/50 error-free, exposing delegation/context overhead under this budget. ReAct completed 41/50, hand graph 37/50. Scam rejection was 1/6 for ReAct, 5/6 for supervisor and 6/6 for hand graph. ReAct's higher completion count therefore includes unsafe synthetic bids; it is not a sufficient reason to ship it.

Failures, latency and cost

Of 48 failures, 39 hit token/step budgets, four timed out, three were provider or structured-output failures and two were other observed stops. Measured successful-call tokens total 1,368,464; 58 native calls had unknown usage. Waits within attempts contribute to latency; between-attempt daily-quota pauses are not fully represented. All three p95 values exceed the declared 60-second analysis target. API billing was not independently verified and remains unknown, not asserted as zero.

40-task tool condition	Tool / argument accuracy	Tokens/task
fine	100% / 100%	617.7
coarse	100% / 100%	988.8
rewritten	100% / 100%	690.6

All 120 GPT-OSS 120B native tool calls were correct on this easy synthetic set. Fine tools used fewer tokens than coarse tools; rewritten descriptions did not improve already-perfect accuracy. No claim of real-task superiority follows.

Evidence: gemini-orchestration-v1/{plan,analysis,progress}.json and runs.jsonl; student-tool-study-v3/{plan,results}.json under evals/results/.

09 / Grounding, human review and X2

The independent judge is Nvidia Nemotron 3.5 Lightning via the fixed OpenRouter free route, rather than either Gemini generator. Its rubric was frozen before human labels were compared. Output-protocol pilots with extra claim indices were rejected, retained separately, and repaired using per-item bounded claim schemas; the support and quality rubric was not tuned to the human answers.

Calibration on 30 human-labelled examples	Observed agreement
Supported / unsupported / non-factual	60%; Cohen kappa 0.403
Quality, 1-5 ordinal scale	46.7% exact; weighted kappa 0.681

Claim agreement is limited. Judge scores below are proxy estimates, not verified truth. One human and purpose-built examples do not establish transfer to full drafts. All available drafts, including later-failing runs, are scored; no-draft runs remain failures in the system metrics. Submitted-only figures are saved separately.

Available drafts	n	Mean quality / 5	Unsupported / factual claims	Drafts flagged
react	40	3.65	19/222	14
supervisor	20	3.75	12/105	6
hand_graph	43	3.88	11/210	8

Blind classmate panel

Two user-supplied reviewers rated twelve anonymized raw Qwen proposal pairs: 13 votes for 4B and 11 for 8B. Reviewers agreed on 11/12 pairs (kappa 0.833). This small convenience panel does not establish a winning model. Twenty-four ratings are twelve repeated jobs, not twenty-four independent jobs. Reviewer identity/independence was not verified; student-2 was reconstructed from the user's statement that only the final choice differed. This panel evaluates the earlier local raw candidates, not current Gemini architecture quality.

X2: randomized proposal opening

The hand-graph cohort was assigned before execution to evidence-first or problem-first openings, 25 jobs each. Intention-to-treat replies were 12/25 (48%) and 17/25 (68%). Evidence-minus-problem difference: -20 percentage points; two-sided Fisher p=0.252, conservative 95% Hoeffding interval [-79.2, +39.2] points. The observed difference is inconclusive. The pooled test ignores assignment strata; per-persona counts are saved. Rule-client outcomes are not real demand.

Evidence: evals/results/judge-calibration-v4/agreement.json; judge-proposals-final/results.json; human-panel-v1/results.json; gemini-orchestration-v1/analysis.json.

10 / Delivery, limitations and reproduction

Requirement	Evidence state
R&D A / B / C	Tool design, five retrieval variants, two open configurations and 150 architecture executions recorded; all negative results remain visible.
Human evaluation and X2	30 calibration labels, two blind panel responses, independent proposal judging and randomized rule-client outcome analysis recorded.
Public delivery	Public repository/article, installable MCP command and Inspector screenshots. Approved proposal/crash, scam, injection and real Critic trace cases verified. Temporary app/HTTP demo is off; E6 uptime is excluded, not satisfied.
Integration gaps	GitHub README/code import and immutable audit snapshots are verified. Real Upwork OAuth/live reads are verified; preview writes are fixture-tested. Final submissions/messages are disabled; production-wide auth limits remain incomplete.
Unmeasured requirements	Paired component-level injection-boundary study is recorded separately. No hidden-test results or observed instructor-policy-change implementation time is claimed.
Video	Genuine three-minute UI recording and 30-second GIF: proposal approval, actual API restart, scam/injection rejection and approved client reply. Fictional account, real hosted inference, one simulator submission receipt.

Engineering finding

The custom graph improves intermediate decision coverage but does not eliminate writer/verifier budget failures. ReAct completes more jobs under this protocol. A reliable approval boundary and durable replay do not imply reliable generation. The next justified work is reducing verifier/context overhead, testing on less templated data, and obtaining stronger human agreement before quality claims.

Reproduce without mixing cohorts

Use https://github.com/raahim-ibrar/GigBidA2 and the current README. Run the download/preprocessing scripts, student_retrieval, student_generation and student_orchestration runners with their frozen plans and source snapshots. Gemini results are the optional closed-model profile; local Qwen and partial Groq runs remain distinct. Run evals.analyse_final_research and evals.analyse_proposal_quality to derive the final tables without new inference. Secrets and raw historical job text are excluded from Git.

AI assistance and responsibility

Codex assisted with implementation, tests, experiment tooling and documentation. The student supplied the calibration labels and reported classmate responses. Fictional sources and outcomes are labelled; no model-generated observation is presented as a human rating. The student must review the work and explain these trade-offs in the viva. Technical narrative: docs/TECHNICAL_WRITEUP.md.

Evidence: docs/{STATUS,RESULTS,VERIFICATION,AI_USAGE,TECHNICAL_WRITEUP}.md; evals/results/.